

INTERNET OF THINGS (IOT) & CYBER-PHYSICAL SYSTEMS (CPS)

The Internet of Things (IoT) refers to a network of physical objects—devices, machines, sensors, vehicles, appliances, and other “things”—that can sense conditions, process data, communicate over networks, and in many cases perform actions. IoT extends computing beyond traditional computers and smartphones into the physical world. A smart thermostat, industrial sensor, connected medical device, agricultural monitor, or vehicle can become part of an IoT system.

A Cyber-Physical System (CPS) goes a step further by tightly integrating computation, communication, and physical processes. A CPS continuously observes a physical environment, processes information, makes decisions, and influences the physical process through actuators. Examples include automated manufacturing lines, smart grids, autonomous vehicles, robotic systems, and intelligent traffic control.

For electronics and computer engineers, IoT and CPS are important because they bring together sensors, embedded processors, communication technologies, software, control, data analytics, and human users.

1. The Basic Architecture of IoT

Although implementations differ, an IoT system commonly contains several interacting layers:

- Physical/Device Layer – sensors, actuators, microcontrollers, embedded processors, RFID devices, cameras, and other connected hardware.
- Connectivity Layer – wired or wireless links such as Ethernet, Wi-Fi, Bluetooth, cellular networks, LPWAN technologies, and other communication systems.
- Edge/Gateway Layer – local devices that collect, filter, translate, analyze, or temporarily store data before sending it to other systems.
- Platform/Cloud Layer – servers and software that store data, manage devices, run analytics, and provide computing services.
- Application Layer – user-facing systems such as dashboards, mobile applications, industrial monitoring tools, smart-home controls, and decision-support systems.

Data generally flows from sensors to processing systems, while commands or control decisions can flow in the opposite direction toward actuators. This creates a feedback loop between the digital and physical worlds.

2. Sensors, Actuators, and Embedded Systems

Sensors

Sensors convert physical phenomena into measurable electrical or digital information. Temperature, pressure, light, motion, humidity, acceleration, distance, sound, and chemical concentration are examples of quantities that can be sensed. Sensor selection requires attention to range, accuracy, resolution, response time, power consumption, calibration, and environmental conditions.

Actuators

Actuators convert electrical or digital commands into physical action. Motors, relays, valves, solenoids, heaters, displays, and robotic mechanisms are common examples. In a CPS, an actuator may change the physical process based on a computed decision.

Embedded Systems

An embedded system is a computing system designed for a specific function within a larger product or process. Microcontrollers and embedded processors often provide the local intelligence in IoT devices. Engineers must consider timing, memory, power, reliability, interfaces, and safety—not simply whether the software produces the correct output.

3. IoT versus Cyber-Physical Systems

IoT and CPS overlap, but they are not identical. IoT emphasizes connected physical objects and the exchange of data across networks. CPS emphasizes the close interaction between computation and physical processes, often including sensing, control, feedback, timing, and safety requirements. An IoT temperature sensor that reports readings to a cloud dashboard is primarily an IoT application. A temperature-control system that senses a process, computes a control action, and automatically regulates a heater is a CPS; if it is network-connected, it can also be considered an IoT-enabled CPS.

4. Communication and Data in IoT/CPS

Communication allows devices to cooperate and allows engineers or users to observe and control systems remotely. Selection of a communication technology depends on range, data rate, latency, reliability, energy consumption, cost, and environmental conditions. Some applications need high bandwidth, while others transmit only small sensor readings but must operate for years on battery power.

IoT systems also generate large quantities of time-stamped data. Data may be filtered at the edge, stored in a local gateway, or sent to cloud services for further analysis. Edge computing is especially useful when a system requires low latency, reduced network traffic, privacy, or continued operation when connectivity is unreliable.

5. IoT/CPS Applications

- Smart homes and buildings – lighting, HVAC, security, energy monitoring, and access control.
- Smart agriculture – soil and weather sensing, automated irrigation, crop monitoring, and livestock tracking.
- Industrial IoT – machine monitoring, predictive maintenance, process optimization, robotics, and production analytics.
- Healthcare – wearable sensors, remote patient monitoring, connected medical equipment, and assisted living.
- Smart transportation – traffic monitoring, fleet management, connected vehicles, and autonomous systems.
- Smart grids – monitoring of generation, transmission, distribution, storage, and energy consumption.
- Disaster and environmental monitoring – flood, fire, weather, air-quality, seismic, and other sensor networks.

6. Security, Safety, and Privacy

Connecting physical devices creates risks that are different from those of ordinary information systems. A compromised IoT device can expose personal information, become part of a larger

cyberattack, or cause physical consequences. Weak passwords, insecure firmware, unprotected communication, outdated software, excessive privileges, and poorly secured interfaces are common risk areas.

- Security by design – consider security from system conception rather than as an afterthought.
- Authentication and authorization – verify devices and users and limit what each is allowed to do.
- Secure communication – protect data in transit and use appropriate cryptographic mechanisms.
- Secure updates – provide mechanisms for authenticated firmware and software updates.
- Data minimization and privacy – collect only what is needed and protect sensitive information.
- Safety and fail-safe behavior – ensure that failures, loss of connectivity, or incorrect data do not create unacceptable physical hazards.

For Philippine applications, engineers should also be familiar with the principles of the Data Privacy Act of 2012 (Republic Act No. 10173), especially when IoT systems collect personal or sensitive personal information.

7. Engineering Design Considerations

Designing an IoT/CPS solution requires more than connecting a sensor to the Internet. Engineers should define the physical problem, identify measurable variables, select suitable sensors and actuators, design the embedded system, choose communications, determine where computation should occur, and establish requirements for reliability, security, maintainability, and safety.

- Power – battery life, energy harvesting, sleep modes, and efficient communications.
- Reliability – component quality, redundancy, fault detection, environmental protection, and recovery.
- Real-time behavior – response time, timing constraints, latency, and deterministic operation.
- Scalability – ability to support additional devices, users, and data without excessive complexity.
- Interoperability – use of appropriate interfaces, protocols, and standards so different devices can work together.
- Maintainability – diagnostics, remote monitoring, logging, and secure updates.
- Safety – identify hazards and define safe states for abnormal conditions.

8. Industry 4.0 and the Future

IoT and CPS are central technologies in Industry 4.0, where connected machines, automation, robotics, analytics, artificial intelligence, and digital twins are integrated into production and services. The trend is moving from isolated automated machines toward intelligent, connected, and increasingly autonomous systems.

Future engineers will increasingly work at the boundary between electronics, computing, communications, control, and data. Emerging approaches such as edge AI, digital twins, 5G/6G connectivity, autonomous systems, and low-power intelligent sensors will expand the

capabilities of IoT/CPS. At the same time, greater connectivity increases the importance of cybersecurity, privacy, resilience, ethics, and responsible engineering.